Indoor Wi-Fi Localization: A Phase 1 Machine Learning Baseline

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Project 9 – Applied Machine Learning

Abstract—Indoor Wi-Fi fingerprinting estimates a device's location from the received signal strengths of nearby access points. This Phase 1 study establishes a reproducible machine-learning baseline on UJIIndoorLoc for building and floor classification and coordinate regression. We removed 637 duplicate training scans, excluded 55 access-point columns constant in training, retained and flagged 224 unusually strong readings, and evaluated five user-disjoint folds. Missing access points were represented consistently, and feature transformations and coordinate scaling were fit within training folds. Nested USERID-grouped tuning compared Random Forest and XGBoost classifiers; k-nearest-neighbor, Random Forest, and XGBoost regressors were evaluated with global, per-building, and per-building-floor routing. Pooled out-of-fold results show 99.34% building accuracy for tuned Random Forest and 77.10% floor accuracy for tuned XGBoost. A kNN per-building route achieved 12.21 m mean position error, 7.52 m median error, and 37.11 m at the 95th percentile. A task-wise RF-building/XGBoost-floor route improved floor classification over RF/RF, but its per-building-floor kNN route had higher mean position error (13.57 m versus 12.80 m). The official test set had already been evaluated with earlier fixed settings, so it is not used to select or claim performance for these tuned models.

Index Terms—indoor localization, Wi-Fi fingerprinting, UJIIndoorLoc, grouped cross-validation, Random Forest, XGBoost, k-nearest neighbors

I. Introduction

Global Navigation Satellite Systems provide limited coverage indoors. Wi-Fi fingerprinting instead maps a scan of received signal strength indicators (RSSI) to a previously observed location. The UJIIndoorLoc benchmark contains scans from multiple buildings and floors, making the task hierarchical: a system must recognize the building and floor as well as estimate two-dimensional coordinates [1]–[3].

This Phase 1 work builds a machine-learning baseline with a shared data and evaluation pipeline. It compares engineered RSSI representations, tree-based classifiers, and coordinate regressors. The validation design groups by user so scans from one person do not appear in both training and validation for an outer fold. This is intended to estimate performance on unseen users more realistically than a random scan split.

II. Related Work

Wi-Fi fingerprinting is commonly treated as a pattern-recognition problem: a measured signal vector is compared with reference fingerprints or mapped to a location by a learned model. Reviews describe distance-based methods such as kNN alongside probabilistic and machine-learning methods, and identify sparsity, device variation, and environmental change as persistent challenges [2]–[4]. Deep-learning surveys motivate shared representations for localization tasks, but the present study evaluates classical models only; multitask learning is reserved for the next project phase [5].

III. Dataset, EDA, and Quality Analysis

UJIIndoorLoc contains 19,937 training scans and 1,111 separately collected official validation scans (21,048 scans in total). Each source row has 529 columns: 520 WAP signal measurements, four targets (BUILDINGID, FLOOR, LONGITUDE, LATITUDE), and five metadata fields (SPACEID, RELATIVEPOSITION, USERID, PHONEID, TIMESTAMP) [1]. The data covers three buildings and 13 building–floor combinations. The official validation file is treated as a held-out test set; model selection uses the training file and saved USERID-grouped folds only. The two source files total approximately 43 MB.

EDA was conducted on the signal matrix and on relevant target and metadata fields, rather than on metadata alone. We checked the distribution of detected AP counts and observed RSSI, class support by building and floor, scan counts by USERID and PHONEID, coordinate maps, and train-to-test signal summaries. Fig. 1 shows the main exploration plots. B2 contributes the largest building-level sample count and is the only building with floor 4; floor support is therefore not uniform. The coordinate map shows spatial clusters associated with buildings and overlapping horizontal positions across floors, which motivates evaluating both floor labels and coordinates. USERID scan counts are strongly uneven (192 to 4,516 scans); PHONEID 14 contributes 4,827 scans across three users. This repeated-user/device structure is a reason to hold out complete users during validation, rather than randomly splitting rows.

The typical training fingerprint detects a small subset of the 520 available WAPs: the mean is 17.88 and median is 17 detected APs per scan. The official test mean and median are 15.59 and 15. Mean observed RSSI is -78.59 dBm in training and -77.45 dBm in the official test. These shifts are descriptive and were not used for fitting, model selection, or hyperparameter tuning. RSSI histograms also expose a small number of near-zero observed values, which

TABLE I
Training data quality and signal-shift checks

Check	Result or action
Exact duplicate training scans	637 removed; first copy retained
Training-constant WAP columns	55 of 520; detected on training only and dropped from both splits
Cleaned training scans	19,300
Observed RSSI in $[-5, 0]$ dBm	224 values, associated with USERIDs 6 and 14; flagged and retained
Detected APs per scan	Train: mean 17.88, median 17; test: mean 15.59, median 15
Mean observed RSSI	Train: -78.59 dBm; test: -77.45 dBm
Official held-out scans	1,111; kept separate from grouped cross-validation

were flagged for audit instead of being removed without source evidence.

The 637 exact duplicate rows were removed from training, leaving 19,300 scans. We found 55 WAP columns constant in cleaned training data and removed those columns from both partitions based only on the training data. We flagged 224 observed RSSI cells in $[-5, 0]$ dBm, associated with USERIDs 6 and 14, but retained them because the evidence did not establish that they were invalid. Five deterministic outer GroupKFold splits were created by USERID. They have zero USERID overlap between each fold's training and validation subsets, and every cleaned training scan appears in exactly one held-out fold. The official 1,111-row file remains outside this development loop.

IV. Features and Models

A. RSSI representation

The raw dataset uses +100 to encode a WAP that was not detected. This is not a physical RSSI; the pipeline maps absence to -110 dBm and carries a binary absence indicator so a missing AP remains distinguishable from a weak observed AP. It compares three signal views: linear-positive $x = (r + 110)/110$, exponential $x = \exp(r/24)$, and a power view based on the normalized positive signal raised to e. Here, r is the observed RSSI in dBm and missing entries map to zero in each view. Per-scan summary features include detected-AP count, strongest RSSI and its AP index, mean and standard deviation of observed signals, and top-three AP information. After 55 constant WAPs are removed, 465 signal values, 465 absence flags, and 11 summary features produce the 941-feature classifier input. The representations are compared on the same USERID-grouped folds; the linear-positive kNN baseline has 20.26 m mean position error, close to the other two views. PCA and variance selection are fitted inside each training fold, then applied unchanged to its validation fold [2], [4]. In the first fold, PCA retained 56.8%, 71.3%, and 87.1% of feature variance with 50, 100, and 200 components, respectively; a zero-variance filter reduced 941 inputs to 811 in that fold.

B. Classification

Random Forest and XGBoost predict BUILDINGID and FLOOR. Their nonlinear tree ensembles can model interactions among access-point readings. Nested tuning selects parameters on inner USERID-grouped splits; outer held-out users produce out-of-fold predictions. Macro-F1 is considered alongside accuracy because floor support is imbalanced. The best task-wise route combines the tuned RF building classifier with tuned XGBoost floor classifier.

C. Coordinate regression and routing

Coordinate models predict LONGITUDE and LATITUDE in metres. We compare kNN, Random Forest, and XGBoost with global, per-building, and per-building-floor routes. Target scaling is fit using the training partition of each fold and predictions are converted back to metres before scoring. End-to-end routes use out-of-fold M3 predictions; oracle routes using true labels are diagnostics only. The primary regression comparison uses nested tuned preprocessing and tuned RF routes. A second route substitutes XGBoost for floor prediction without retuning the regressors.

V. Experimental Setup

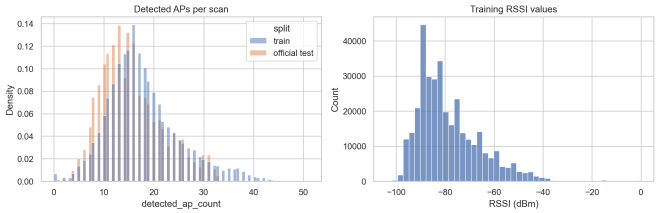
All models use the same five outer GroupKFold splits grouped by USERID. Feature transforms, variance selectors, and coordinate target scalers are fit on the relevant training portion only. Classifier tuning uses nested grouped validation and macro-F1 selection. Regressor tuning uses inner grouped validation and compares coordinate error. We report pooled out-of-fold metrics across the 19,300 training scans. The official test file was evaluated earlier using fixed settings; therefore, its saved numbers are not treated as validation of the tuned pipeline.

Metrics include building accuracy, overall floor accuracy, floor accuracy conditional on a correct building, per-axis coordinate MAE, Euclidean position-error mean/median/95th percentile, hierarchical error breakdown, and an optional EvAAL-style penalized score with 50 m and 4 m penalties for building and floor errors.

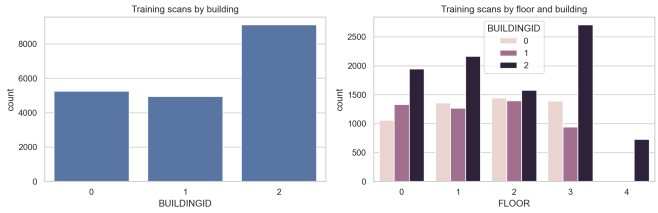
VI. Results

A. Classification

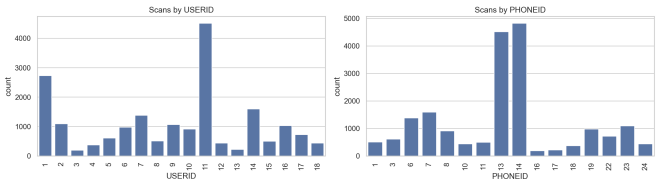
Table II shows that building recognition is substantially easier than floor recognition. Tuned RF achieves 99.34% pooled building accuracy. Tuned XGBoost gives the highest pooled floor accuracy at 77.10%; its outer-fold mean macro-F1 is 0.743. Floor 4 has only 727 out-of-fold scans. Its recall is 9.2% for RF and 23.9% for XGBoost, so overall accuracy alone hides a difficult minority class (Fig. 3).



(a) Detected AP counts and observed training RSSI



(b) Training scan support by building and floor



(c) Scan counts by user and phone



(d) Coordinate distributions by building and floor

Fig. 1. Exploratory analysis of signal coverage, label support, acquisition entities, and coordinate structure.

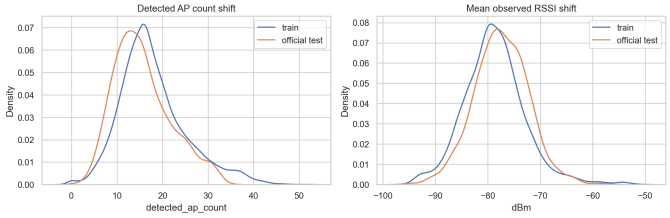


Fig. 2. Train and official-test distributions for detected AP count and mean observed RSSI. The official test is descriptive only for Phase 1 model selection.

TABLE II
Tuned classifier pooled out-of-fold results

Classifier	Building (%)	Floor (%)	Floor | building correct (%)
RF / RF	99.34	74.75	74.98
XGBoost / XGBoost	96.87	77.10	78.71
RF / XGBoost	99.34	77.10	77.52

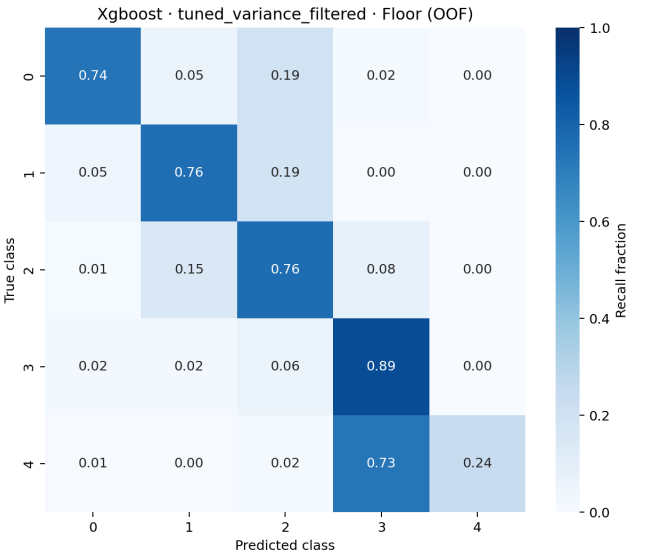


Fig. 3. Pooled out-of-fold floor confusion matrix for tuned XGBoost. Floor 4 is the least-supported class.

B. Coordinate localization

Table III gives the best kNN routes. Per-building kNN has the lowest mean position error in the primary RF/RF route (12.21 m). Per-building-floor routing gives 12.80 m. Substituting XGBoost for the floor classifier improves classification accuracy but the corresponding floor-routed kNN result is 13.57 m. Better floor accuracy therefore did not translate into lower coordinate error in this experiment. The global kNN route has 21.01 m mean error, showing the benefit of separating building spaces.

TABLE III
kNN pooled out-of-fold localization results

M3 route	M4 route	Mean (m)	Median (m)	P95 (m)	Score (m)
RF/RF	Global	21.01	8.85	92.47	22.35
RF/RF	Per-building	12.21	7.52	37.11	13.55
RF/RF	Per-building-floor	12.80	7.65	41.63	14.14
RF/XGB	Per-building-floor	13.57	7.65	46.97	14.82

VII. Discussion

The results establish a reproducible Phase 1 baseline and expose the main weakness: floor predictions, especially the least-supported floor, remain less reliable than building predictions. The analysis also shows why classification and localization scores should be examined together. Changing the floor classifier can improve label accuracy without improving the coordinate route, because an incorrect floor can send a scan to a different specialized regressor and the coordinate models are trained on

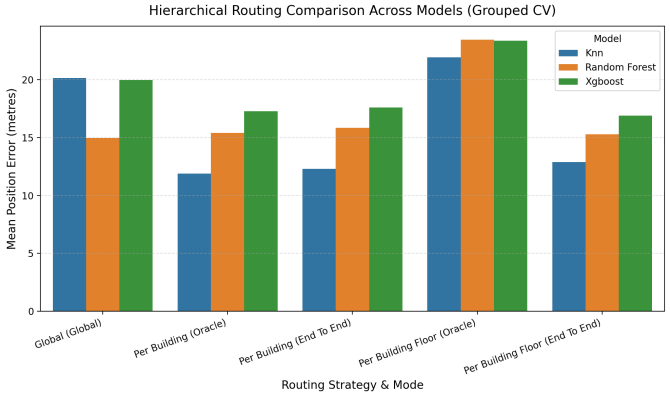


Fig. 4. Oracle and end-to-end routing comparison using grouped-CV regressors.

different partitions.

The user-grouped design and fold-local fitting reduce leakage risk, but the study has limits. The five outer folds are the main evidence for tuned models; the prior official-test run used fixed settings and has already consumed the test set for this project. The current results should therefore not be described as a fresh tuned-model test evaluation. The analysis does not yet include multiple random seeds, paired confidence intervals, calibration testing, or a worst-case error review. Those are additional validation and Phase 2 tasks.

VIII. Conclusion

This Phase 1 implementation provides data-quality evidence, fold-safe feature processing, nested grouped classification and regression comparisons, and a shared evaluation table. RF is the strongest building classifier in the reported comparison, XGBoost gives the strongest floor accuracy, and kNN with per-building routing gives the lowest pooled mean coordinate error. These results form the classical baseline for Phase 2, where a shared multitask network and a justified hybrid can be evaluated on the same grouped splits.

Contribution Summary

Contributions: M1 – data and EDA; M2 – features and representation; M3 – classification; M4 – regression and routing; M5 – evaluation and report integration.

References

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